

REQUIRED COMPETENCIES



BASIC COMPETENCY

Basic competencies are foundational skills, knowledge, and work attitudes that apply to all occupations and industries. They are not technical; instead, they focus on how a person behaves, communicates, and works with others.



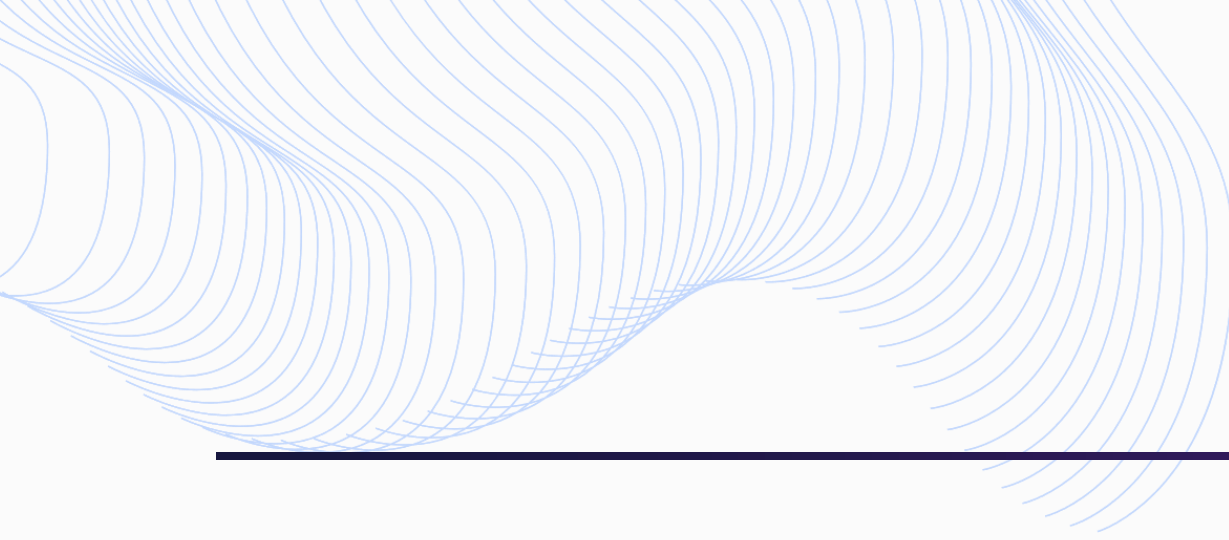
COMMON COMPETENCY

common competencies ensure that trainees: Use standard industry tools and equipment, Follow workplace procedures and standards, Apply quality, safety, and documentation practices relevant to the industry
They are more technical than basic competencies but not job-specific.



CORE COMPETENCY

Core competencies are the specific technical skills, knowledge, and attitudes required to perform the major functions of a particular qualification or job role. These competencies define the qualification.



**INTRODUCTION TO
CYBER**

**THREAT MONITORING
* LVL 1 ***

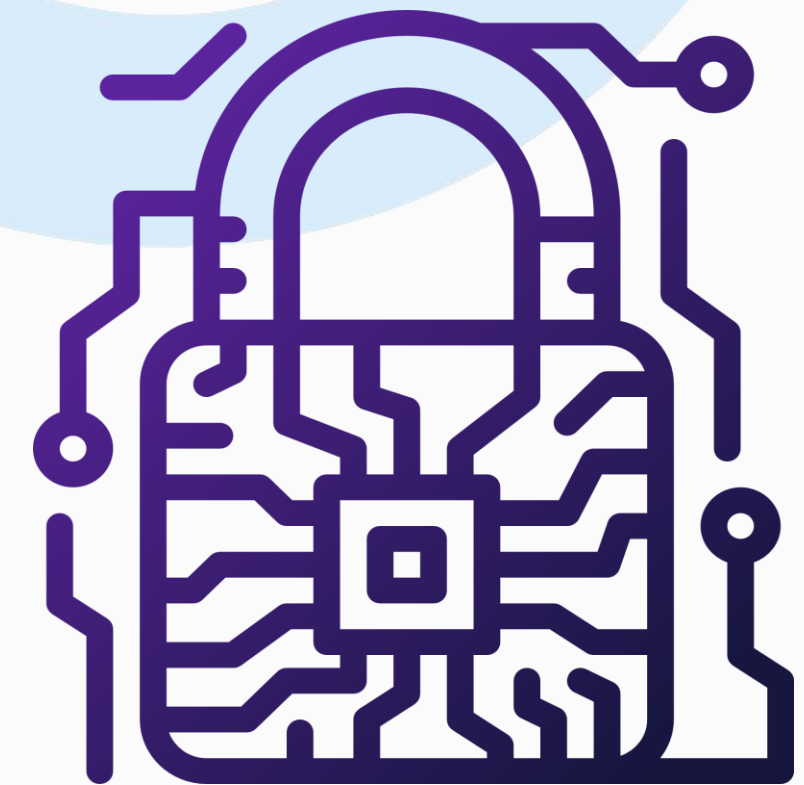
INTRODUCTION

TOPIC 2: Understanding what Cyber Threat Monitoring is

is the continuous process of watching, detecting, and analyzing potential digital threats that could compromise an organization's systems, networks, or data.

It's about **spotting cyber attacks early** or even before they happen so organizations can **prevent damage, data loss, or downtime**.

- **Watching systems, networks, and logs**
- **Detecting suspicious activity**
- **Responding before damage happens**



LEARN THE FOUNDATION

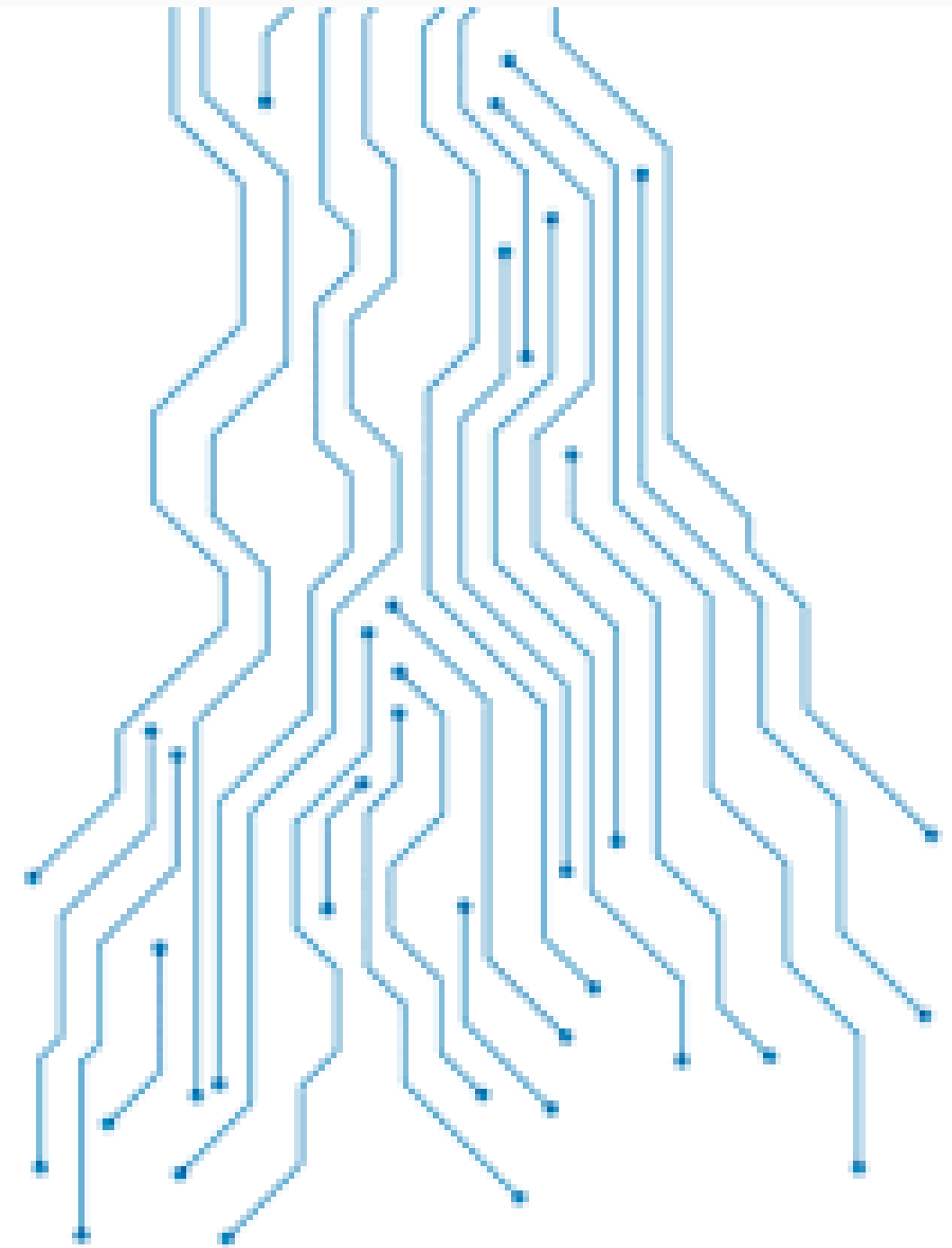
Before tools, you need core knowledge:

a. Networking Basics

You should understand:

- **IP Address**
- **DNS**
- **Ports (like 80, 443)**
- **TCP vs UDP**
- **How data travels**

If you skip this, logs won't make sense.



LEARN THE FOUNDATION

b. Operating Systems

Focus on

- Windows (Event Viewer, logs)
- Linux (basic commands like grep, cat, tail)

c. Basic Cybersecurity Concepts

Learn

- Malware
- Phishing
- Brute force attacks
- Firewall & IDS/IPS
- Logs and alerts



LEARN THE KEY TOOLS

Beginner-friendly tools used in real jobs:

SIEM Tools (Very Important)

SIEM = Security Information and Event Management

Start with

- **Splunk** (best beginner-friendly)
- **Wazuh** (free + powerful)

These tools collect logs and show alerts.



LEARN THE KEY TOOLS

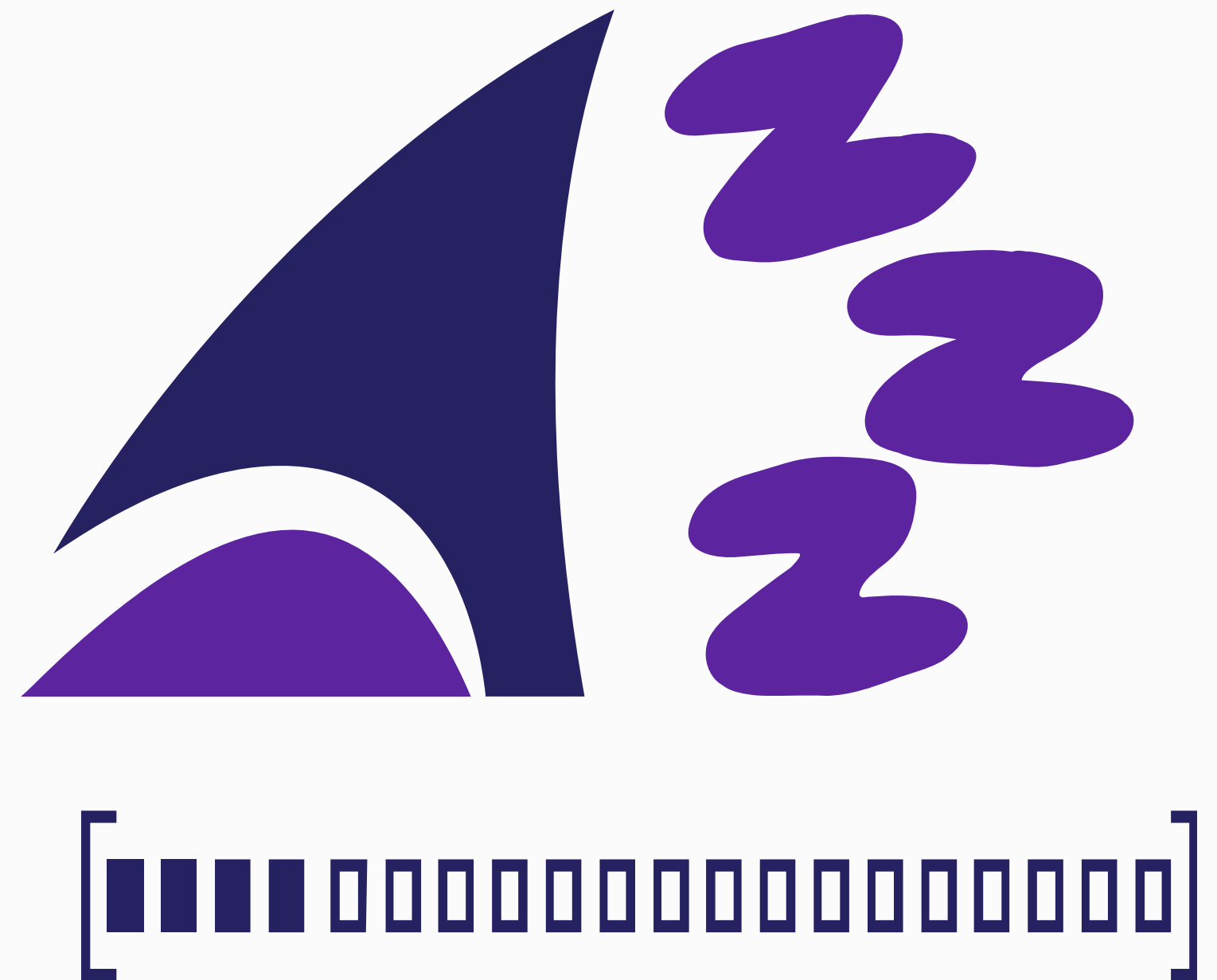
Network Monitoring Tools

- **Wireshark**
- **Zeek**

These let you see real network traffic.

Log Analysis

- Learn how to read logs:
 - Failed login attempts
 - Suspicious IPs
 - Unusual activity



PRACTICE HANDS-ON

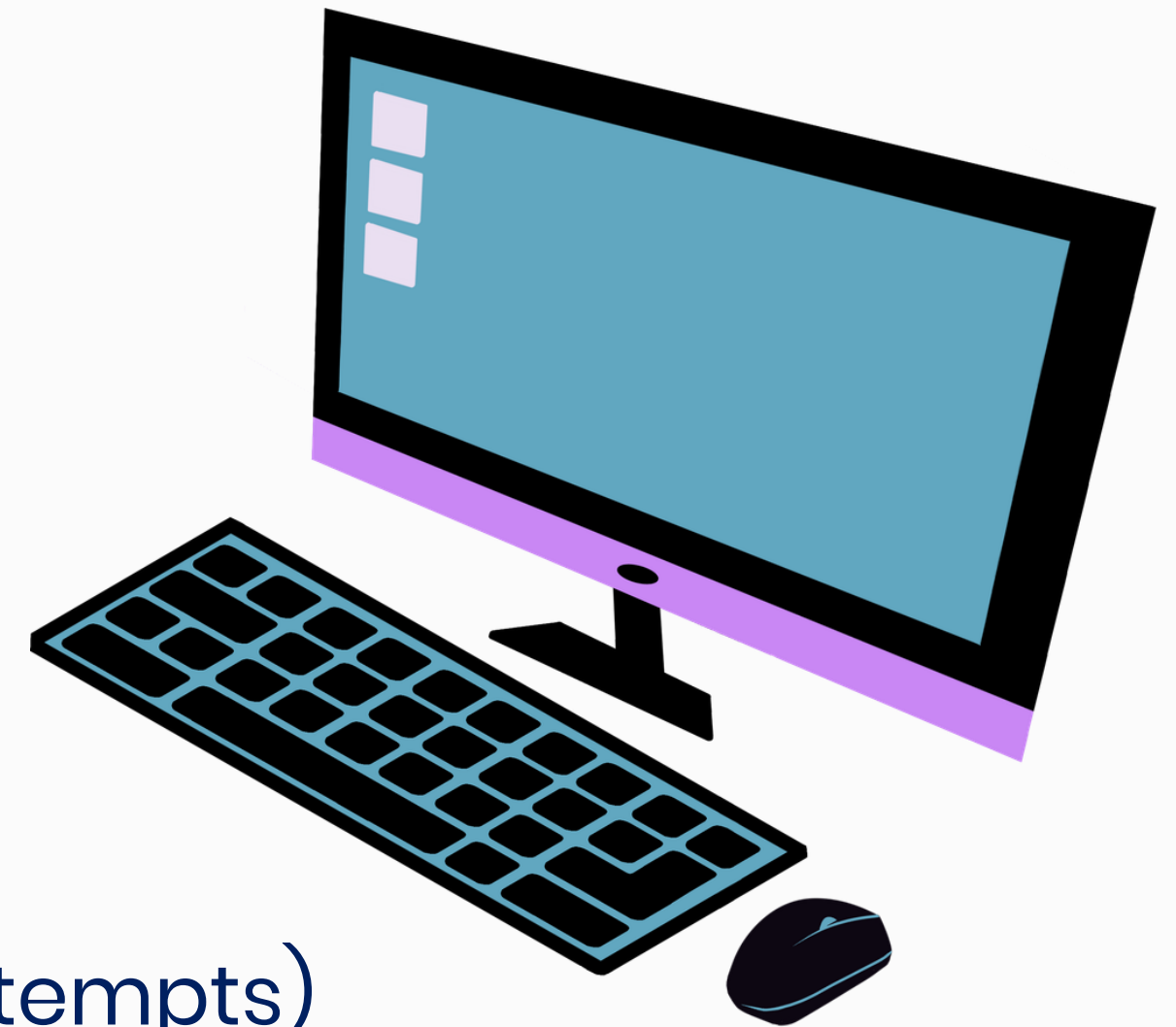
Build your own lab.

Setup your mini cyber lab!

- Install **VirtualBox**
- Create:
 - **1 Windows machine**
 - **1 Linux machine**

Then

- Install **Wazuh**
- **Generate fake attacks** (like wrong password attempts)
- **Observe logs**

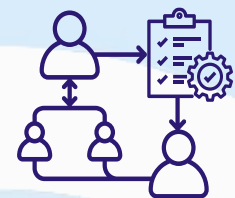


LEARN THREAT DETECTION

Start detecting



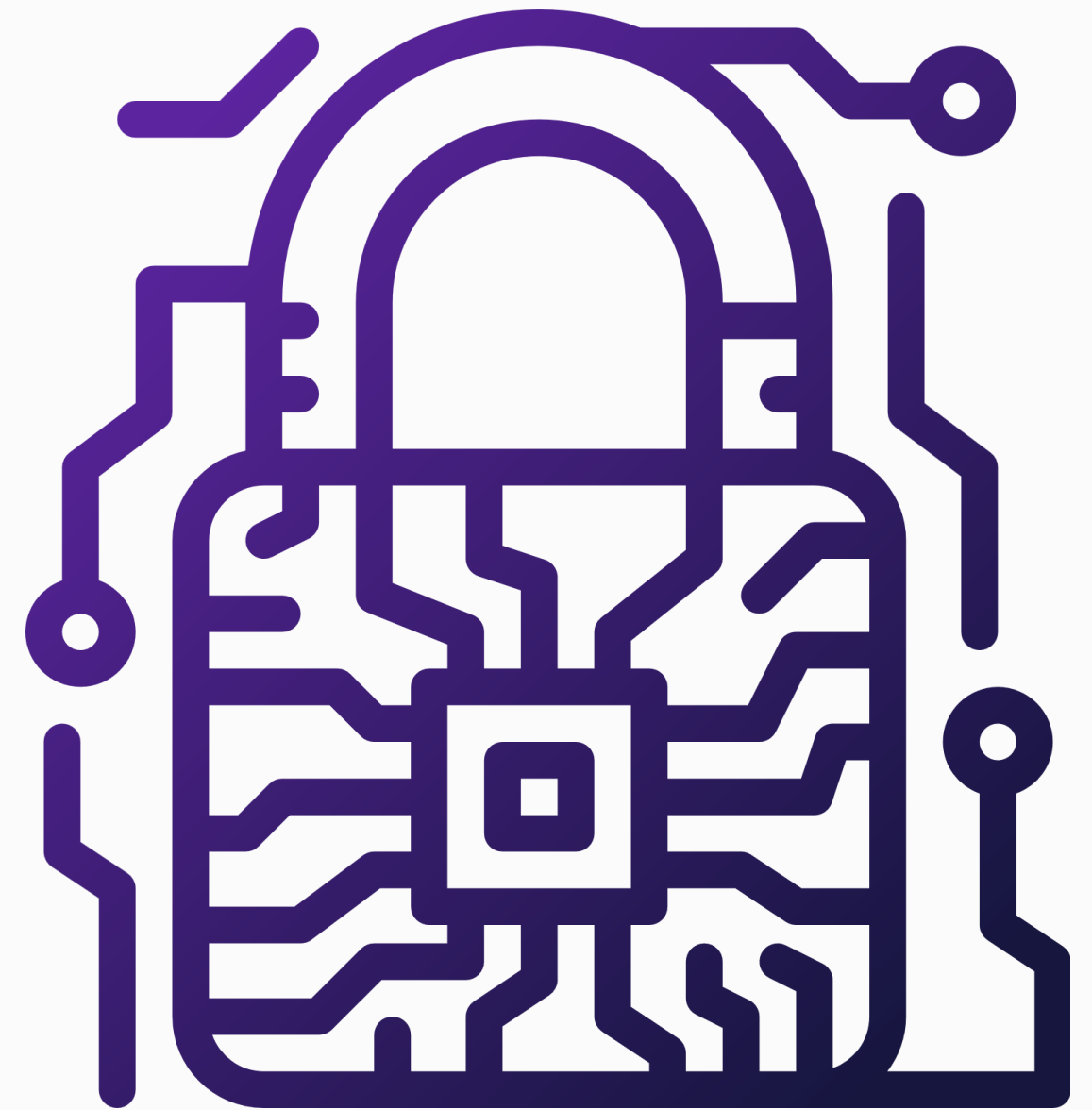
Brute force login attempts



Suspicious IP connections



Malware behavior patterns



FOLLOW REAL WORLD FRAMEWORKS



Later, learn

- **MITRE ATT&CK**

This shows how attackers behave in real life.



Understand the
**DISCIPLINES OF
CYBERSECURITY**



What core of cybersecurity qualifications identifies system vulnerabilities, conducting penetration testing, and adhering to strict legal and ethical guidelines?

- A. CYBER THREAT MONITORING**
- B. CYBER THREAT MITIGATION**
- C. CYBER RISK ASSESSMENT**
- D. ETHICAL HACKING**



What core of cybersecurity qualifications identifies system vulnerabilities, conducting penetration testing, and adhering to strict legal and ethical guidelines?

D. ETHICAL HACKING





What core of cybersecurity qualifications focuses on reporting cyber threats and conducting vulnerability scanning of assets?

- A. CYBER THREAT MONITORING**
- B. CYBER THREAT MITIGATION**
- C. CYBER RISK ASSESSMENT**
- D. ETHICAL HACKING**



What core of cybersecurity qualifications identifies system vulnerabilities, conducting penetration testing, and adhering to strict legal and ethical guidelines?

A. CYBER THREAT MONITORING





Is a unique numerical label assigned to every device connected to a network, acting as a digital location for sending and receiving data.

A. MAC ADDRESS

B. NUMERICAL NETWORK ADDRESS

C. IP ADDRESS

D. DIGITAL UNIQUE ADDRESS



Is a unique numerical label assigned to every device connected to a network, acting as a digital location for sending and receiving data.

C. IP ADDRESS





4 PIC 1 WORD

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Z V C I X K L P A T
M J F L B H R S N G



ANSWER!

M O N I T O R I N G

Cybersecurity protects internet-connected systems such as hardware, software, and data from cyber threats and attacks. Individuals and enterprises use the practice to protect against unauthorized access to data centers and other computerized systems.

Cyber security is a discipline that covers how to defend devices and services from electronic attacks by nefarious actors such as hackers, spammers, and cybercriminals.

Why cyber security matters?



Data protection



Privacy safety

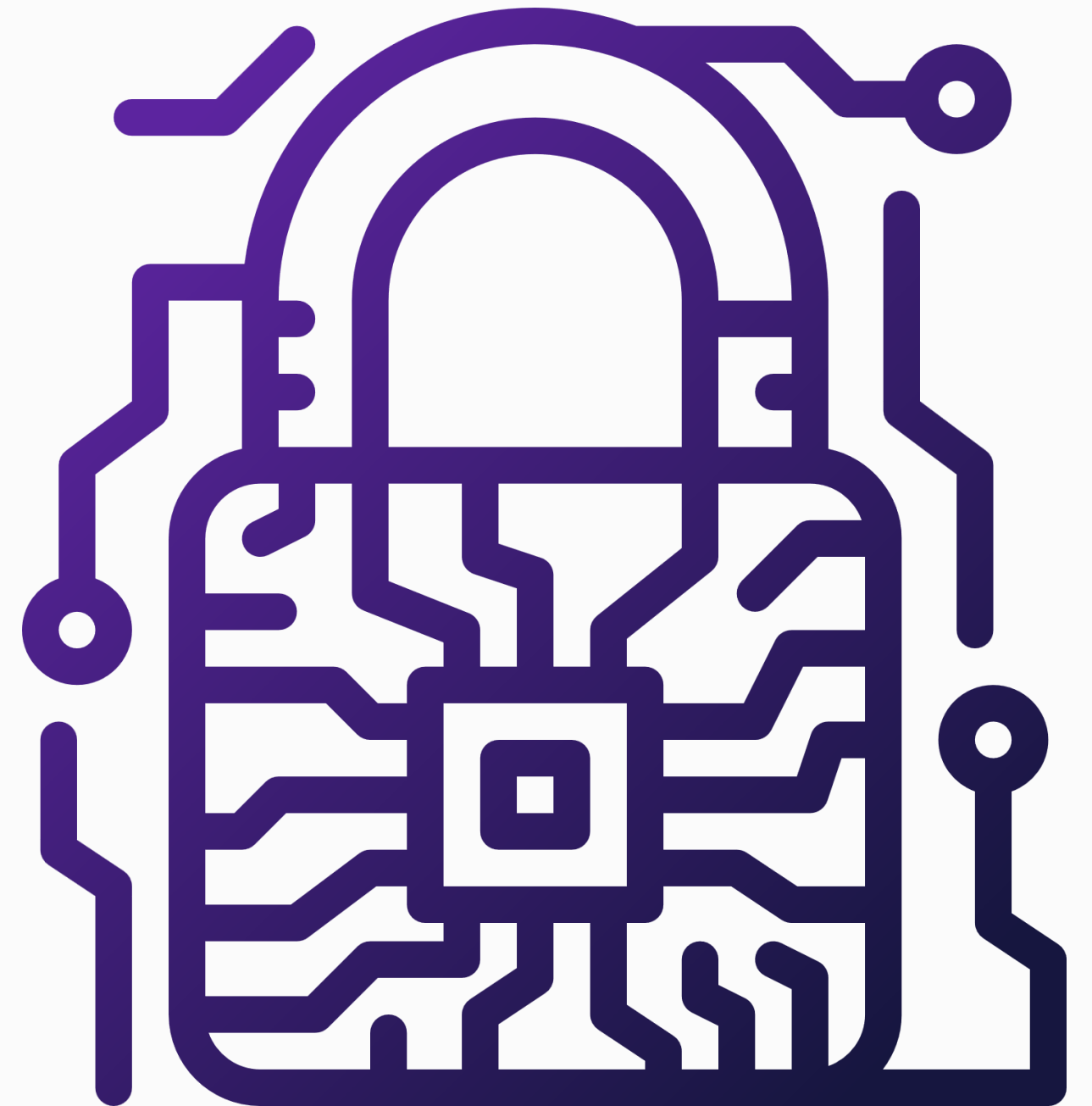


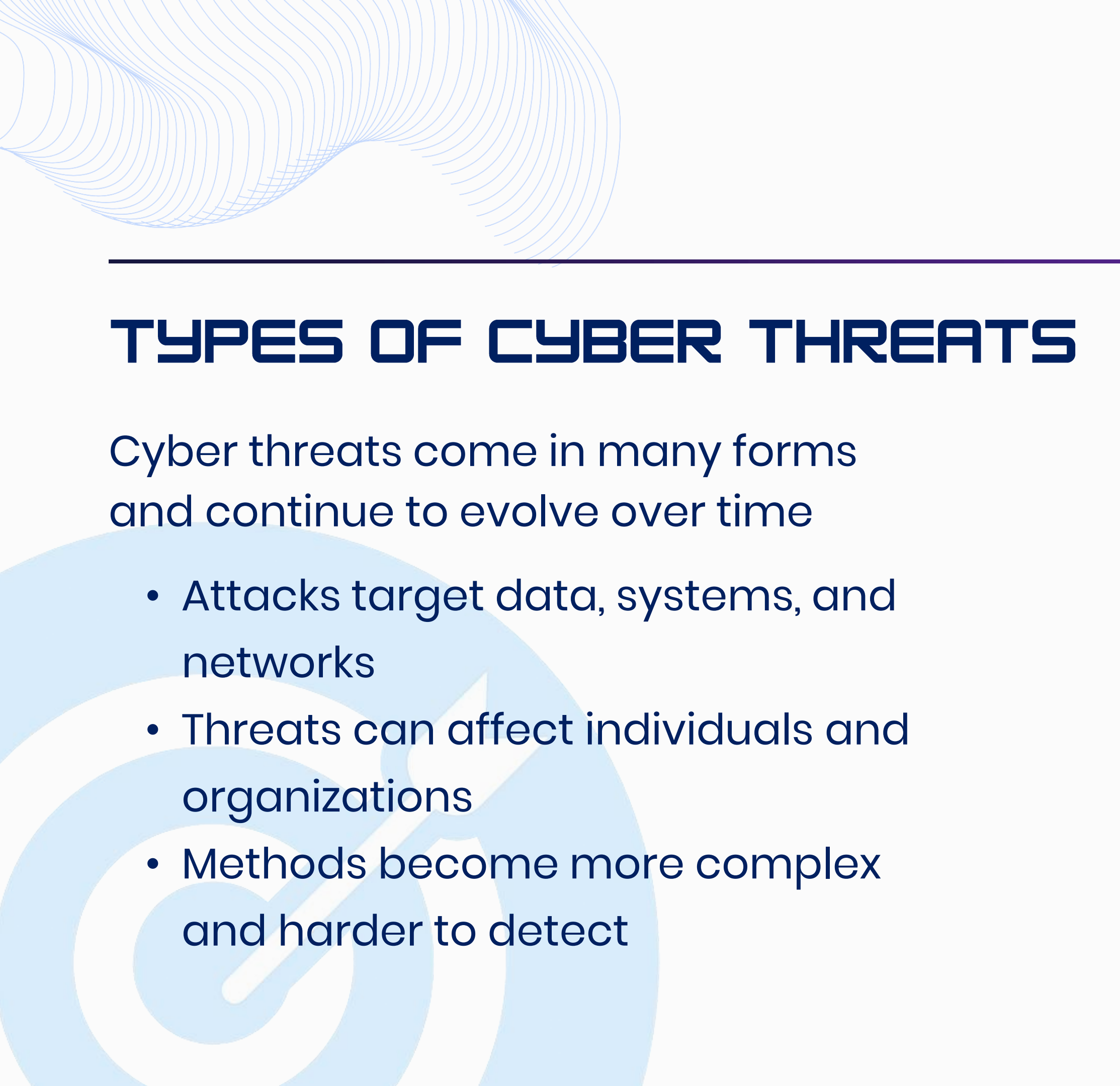
Business continuity

CYBER THREATS?

These are popularly known as **cyber-attacks or hacks**. A cyber threat is an **attempt to expose, alter, disable, destroy, steal, or gain unauthorized access** to or make unauthorized use of an asset.

It refers to anything that has the **potential to cause serious harm to a computer system**. A cyberthreat is something that may or may not happen, but has the potential to cause serious damage.





TYPES OF CYBER THREATS

Cyber threats come in many forms and continue to evolve over time

- Attacks target data, systems, and networks
- Threats can affect individuals and organizations
- Methods become more complex and harder to detect





MALWARE

This attack involves the creation and distribution of malware. Examples of malware are viruses, trojan horses, network worms, ransomware, etc.

RANSOMWARE

is a malware that encrypts a victim's data, files, or devices, rendering them inaccessible, and demands payment (usually cryptocurrency) for the decryption key. It causes significant operational disruption and data loss, often using "double extortion"—stealing data before encrypting it to threaten leaks.

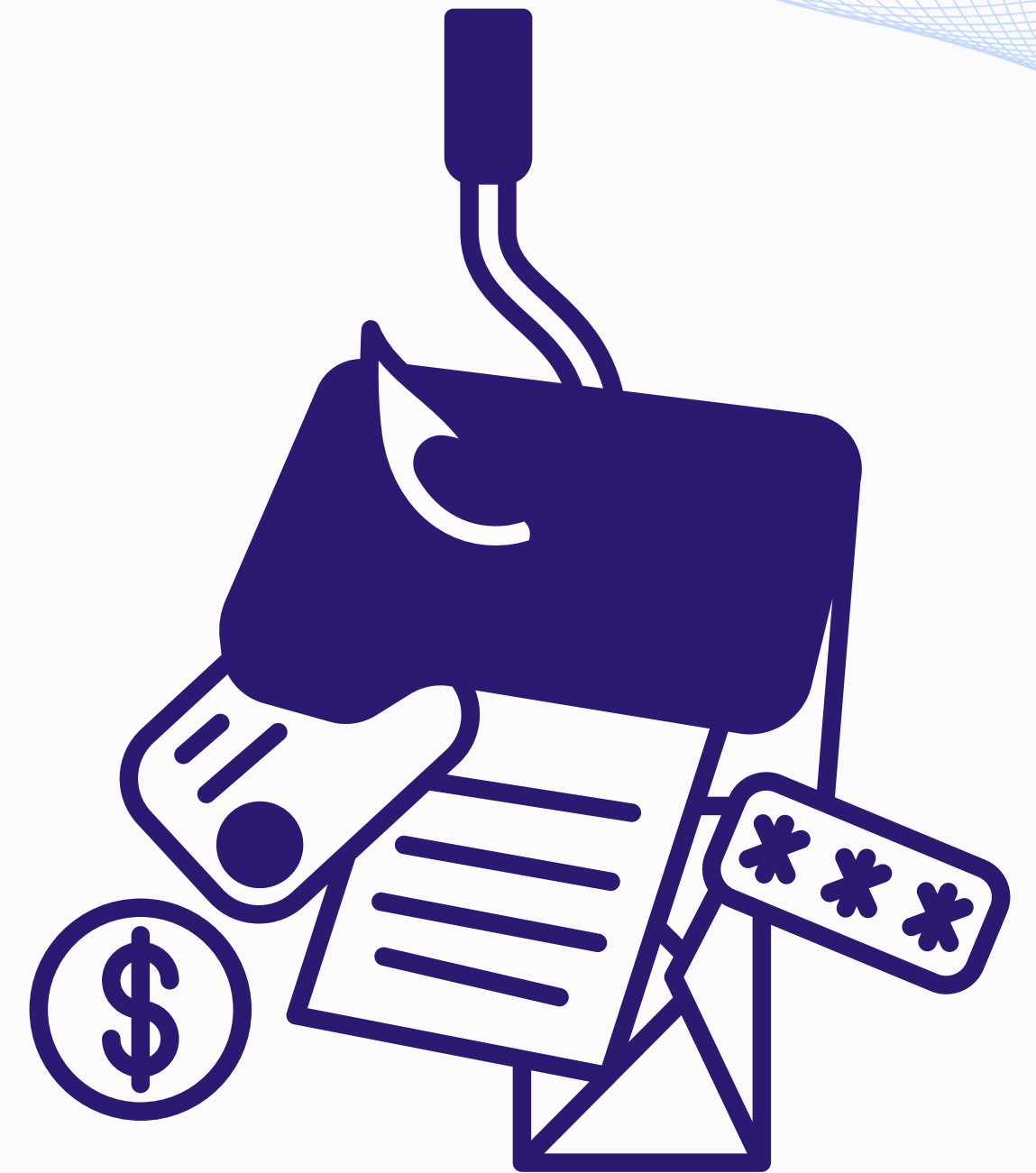
PHISHING

This attack involves an attacker cloning the login page of a website and sharing it with victims in an attempt to steal their login credentials.

SQL/NOSQL INJECTION

This attack is targeted at databases. They attempt to steal or clone the database content of a website or app if the queries constructed from user input are not properly sanitized.

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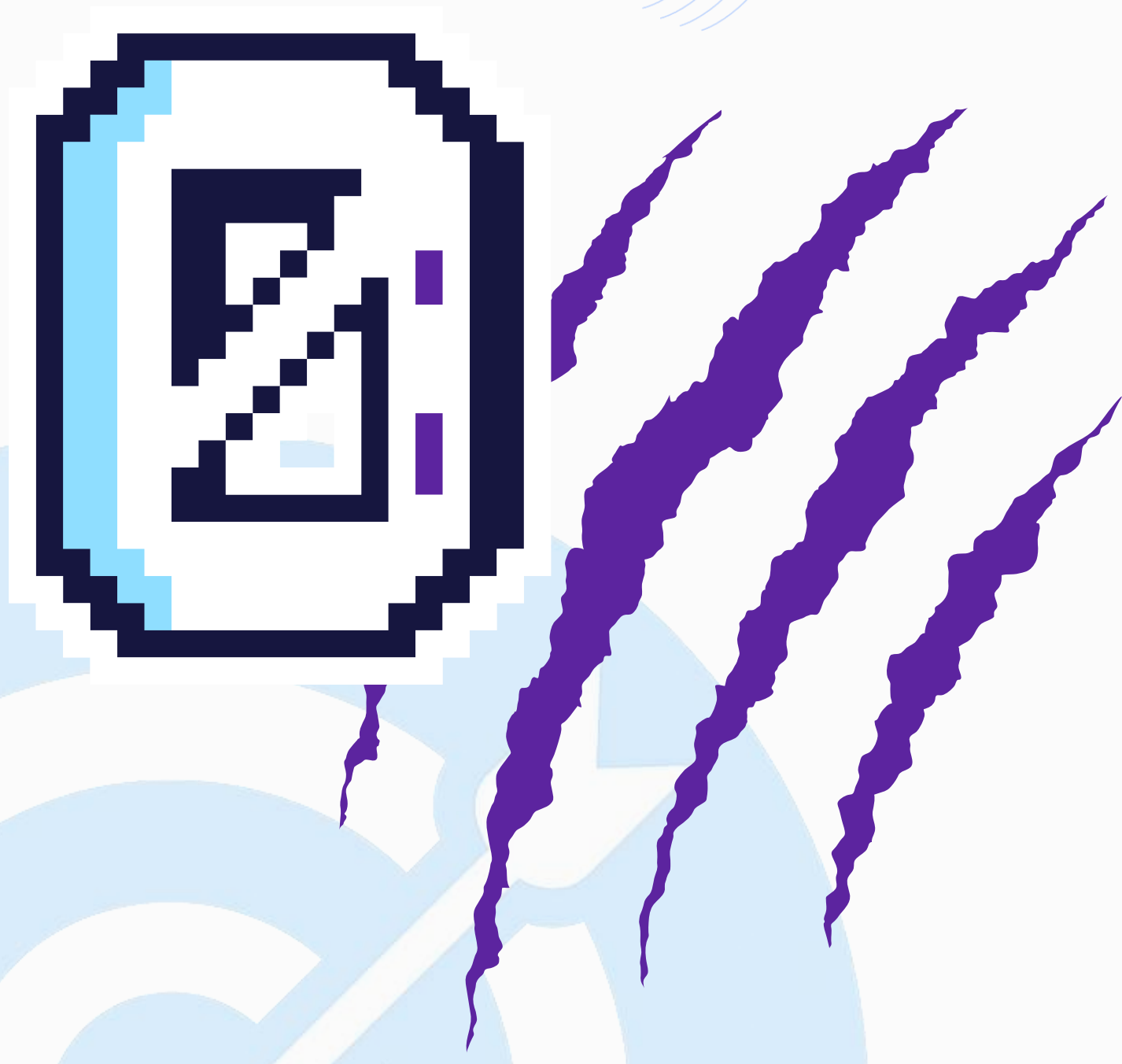


CROSS-SITE SCRIPTING ATTACKS

This attack involves embedding malicious JavaScript code on websites that do not sanitize user input before rendering. Hackers can use them to steal sensitive data from users and send it to an attacker.

BRUTE-FORCE ATTACKS

This attack involves a user trying every possible combination of passwords for a user account on a website.



MITM

Man-in-the-middle Attacks, this attack is typical in networks and occurs when an attacker intercepts requests sent outside a network. An example of this is an attacker seeing form data submitted on websites and even changing their values (the data can include passwords and credit card information).

ZERO-DAY EXPLOITS

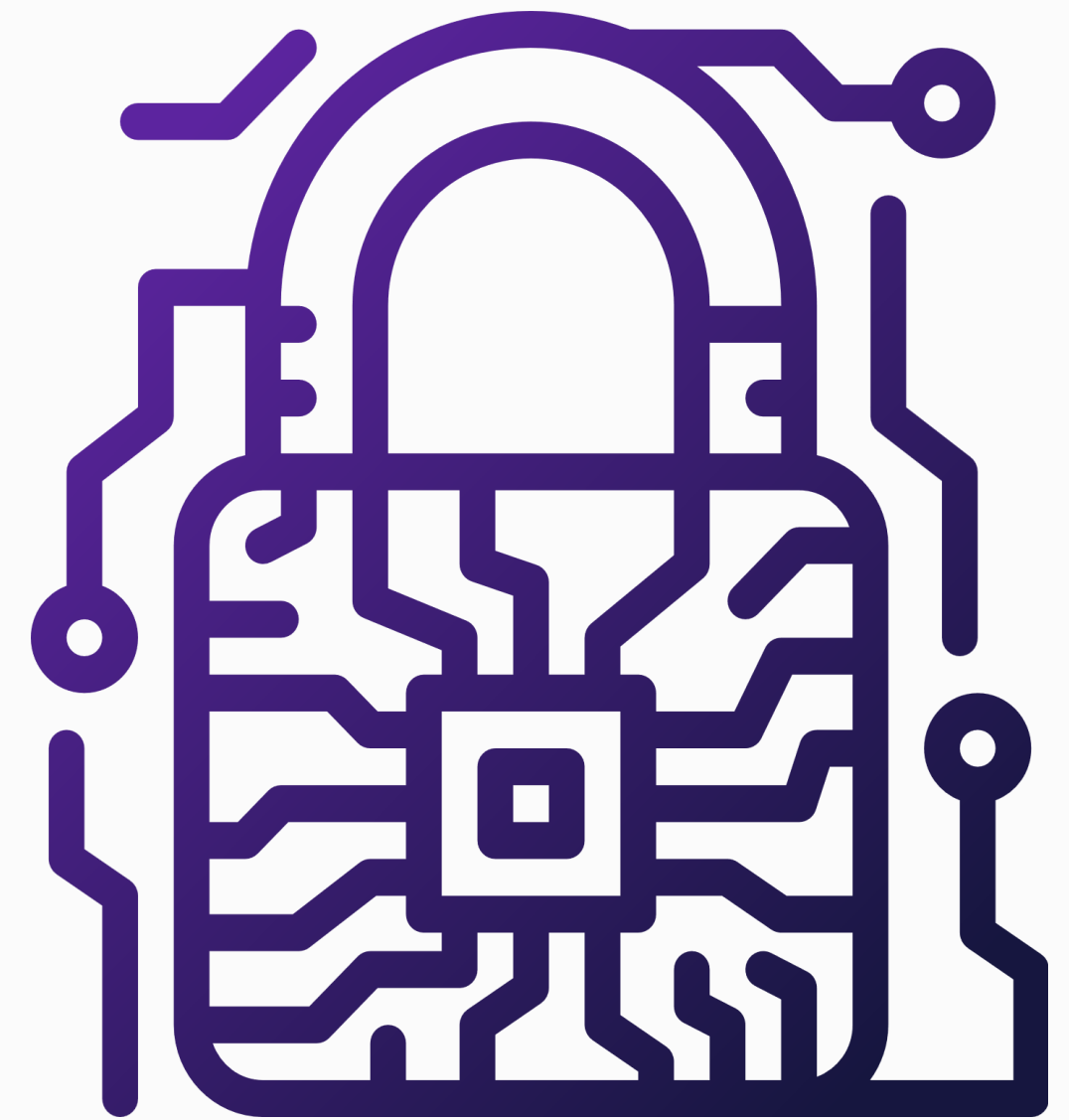
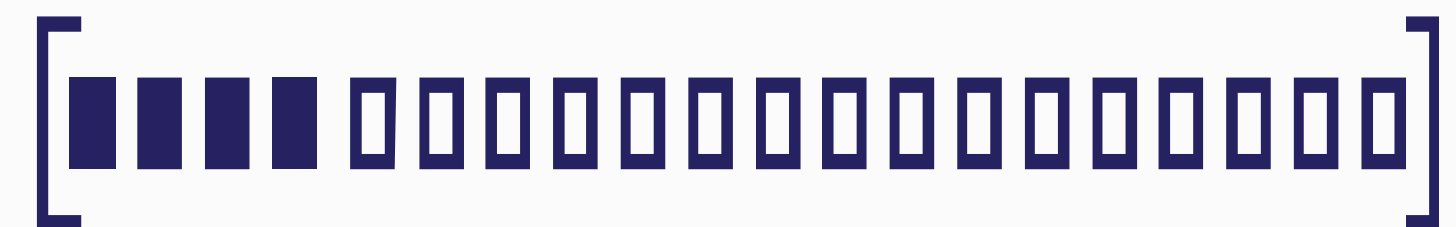
This attack is a newly discovered or developed vulnerability for a system, platform, or application that the creator of the software has not had time to patch or fix.

SOCIAL ENGINEERING ATTACKS

This attack involves an attacker attempting to get access to a system via the users themselves.

CROSS-SITE REQUEST FORGERY ATTACKS

This attack allows an attacker to perform actions on websites that a user is authenticated on without their consent or knowledge.

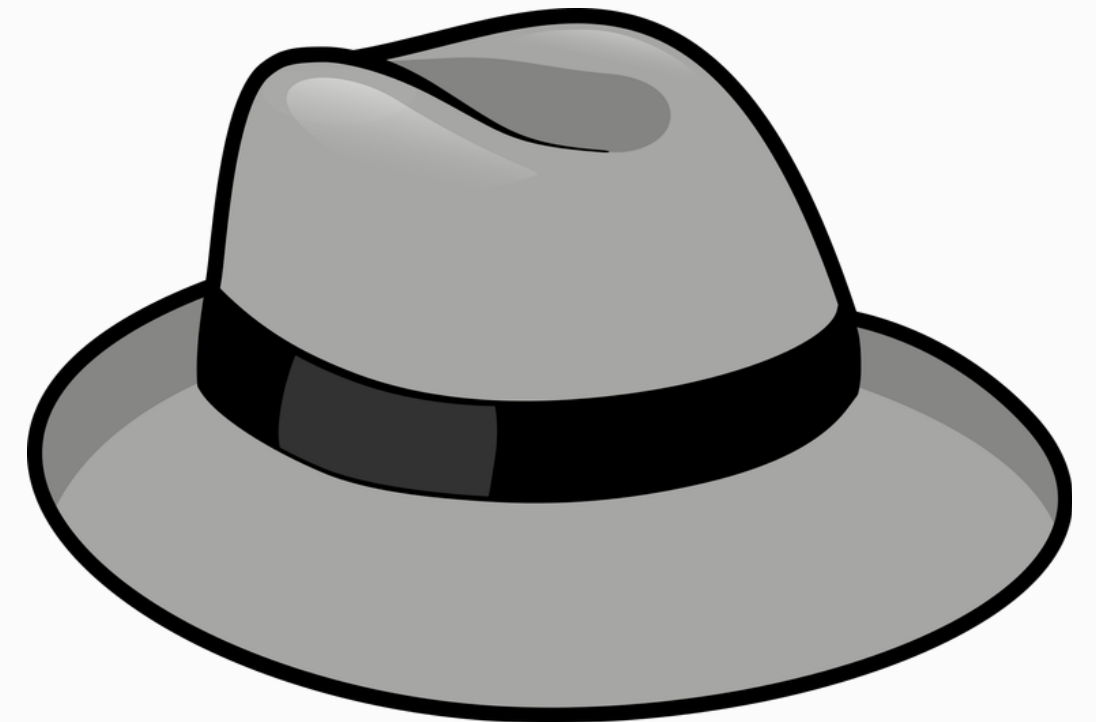
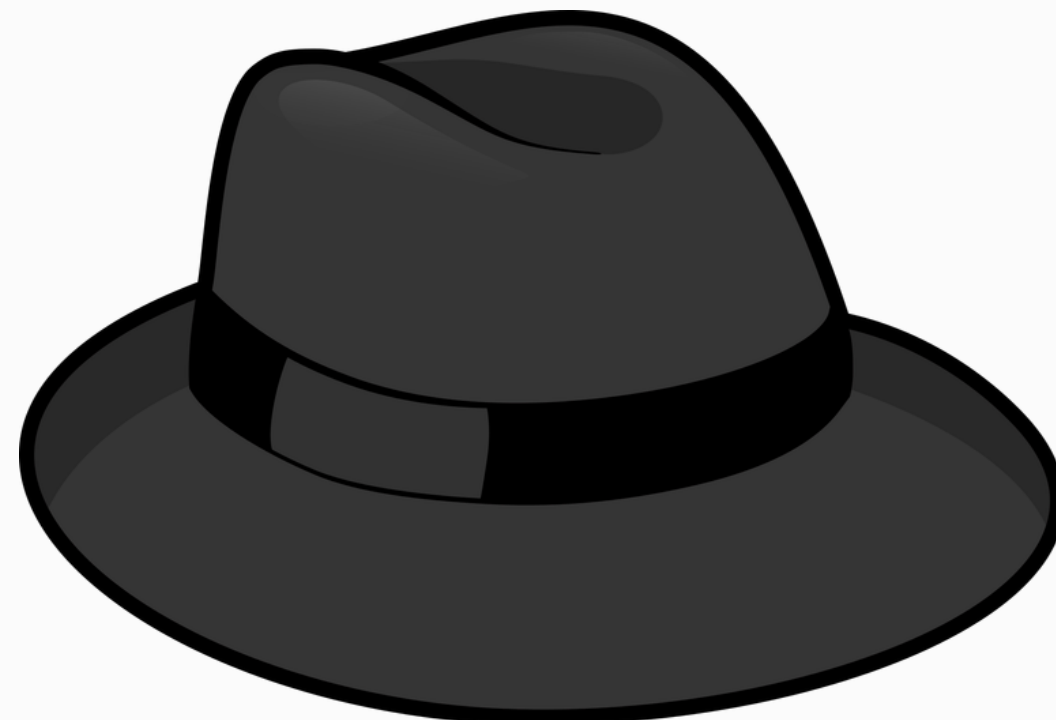
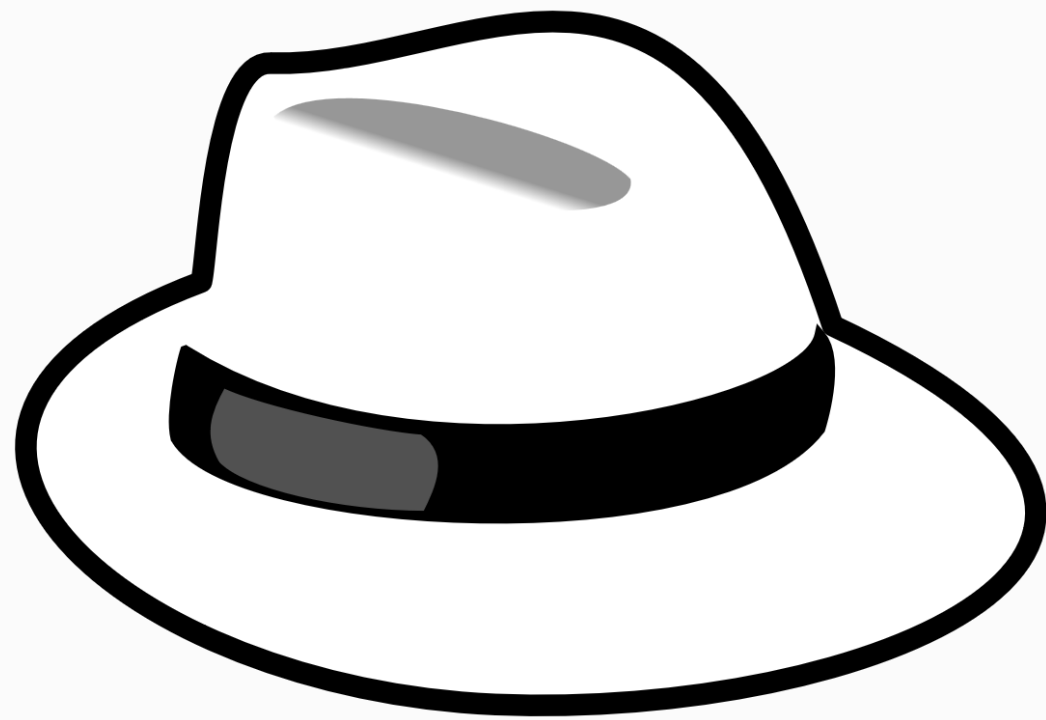


INSIDER THREATS

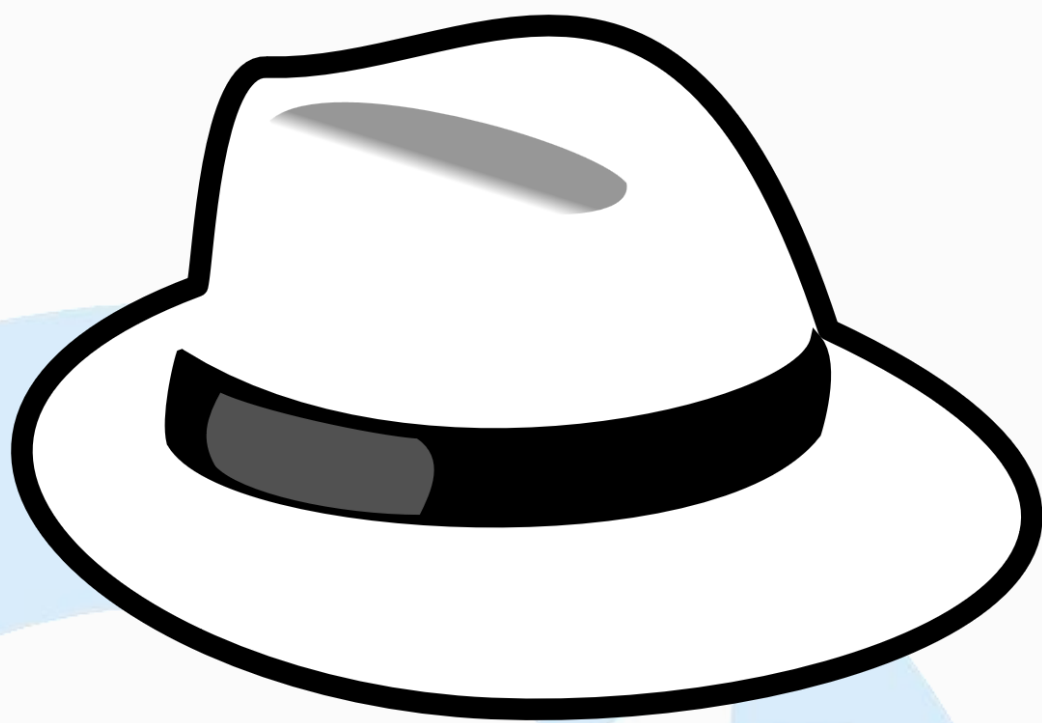
This form of attack involves attacking a system with inside help. This could be an employee who decides to destroy a system and not an unknown outside attacker.



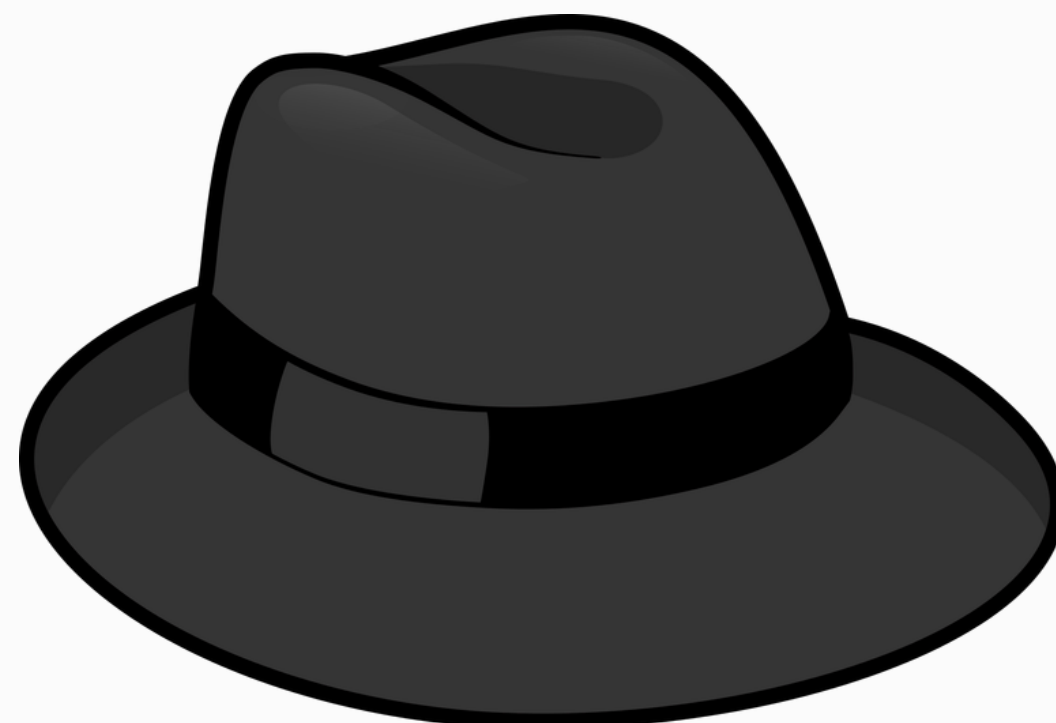
IDENTIFY THE FOLLOWING IMAGES



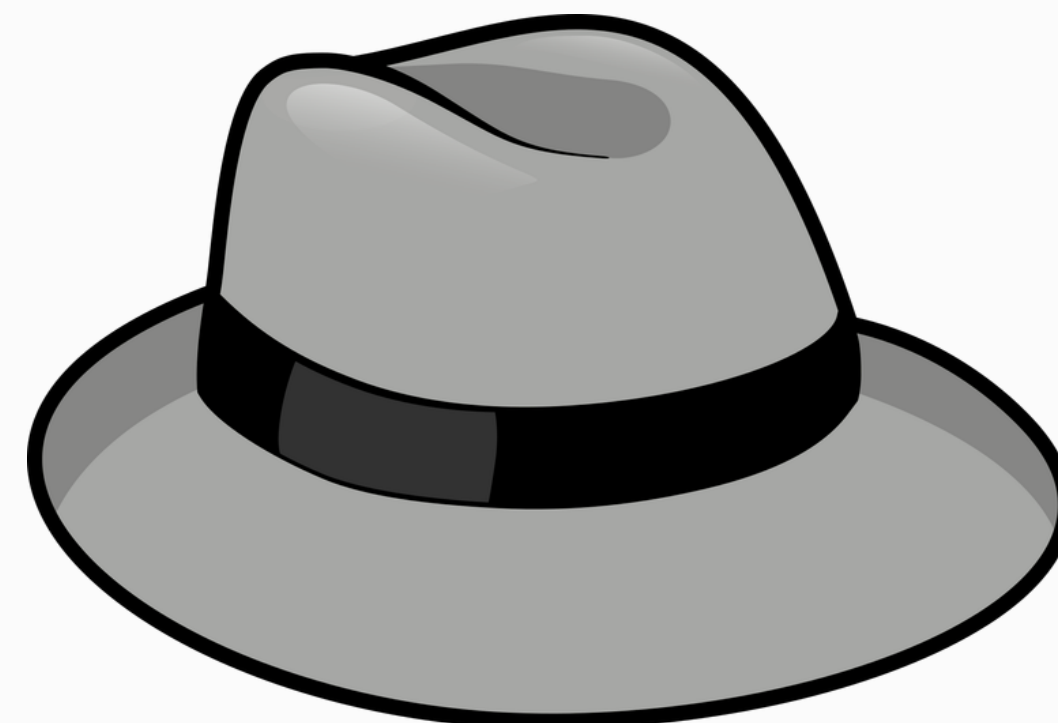
ANSWER!



White Hat



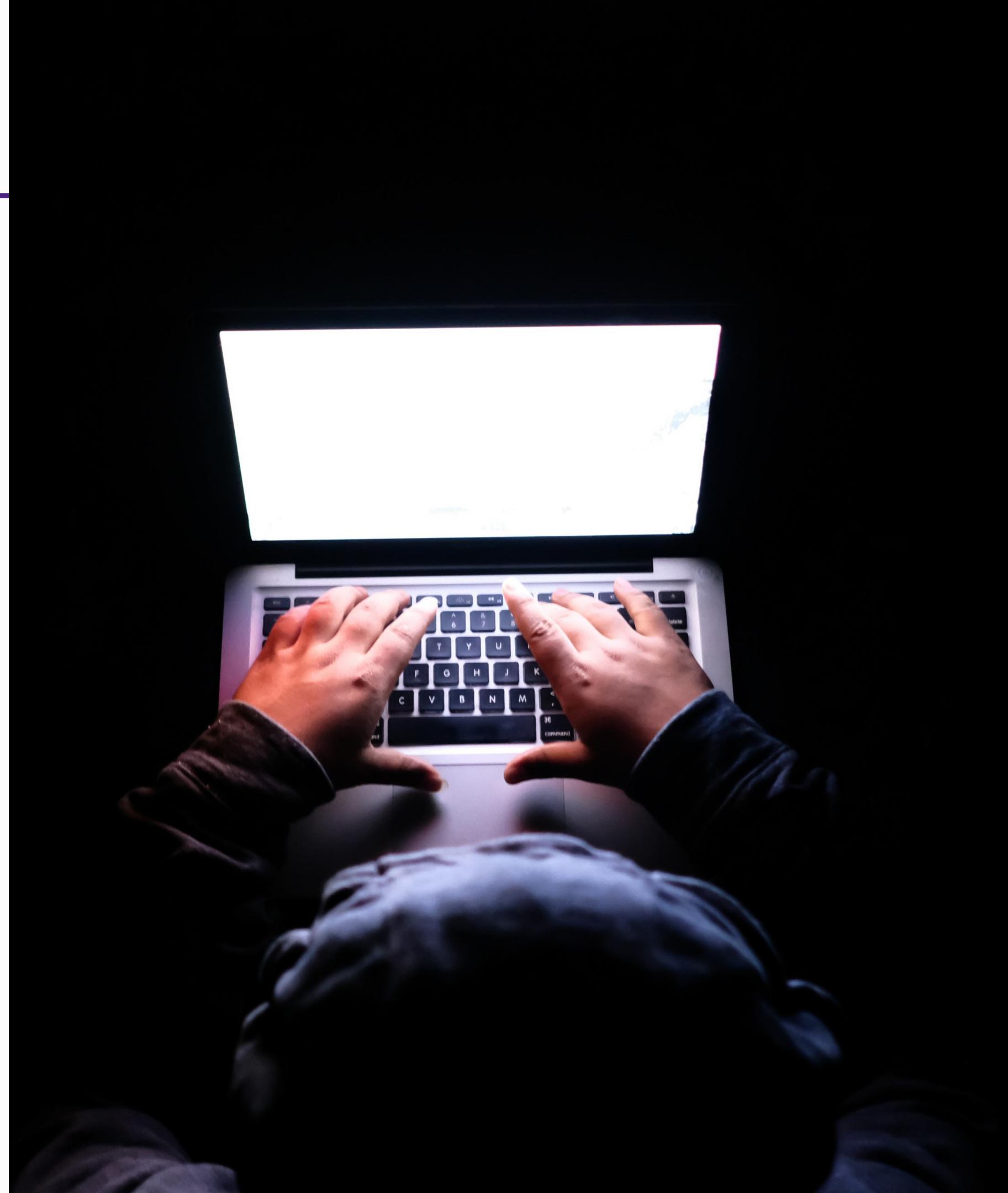
Black Hat



Gray Hat

GUESS THE WORD!

refers to activities that seek to compromise digital devices, such as computers, smartphones, tablets, and even entire networks.

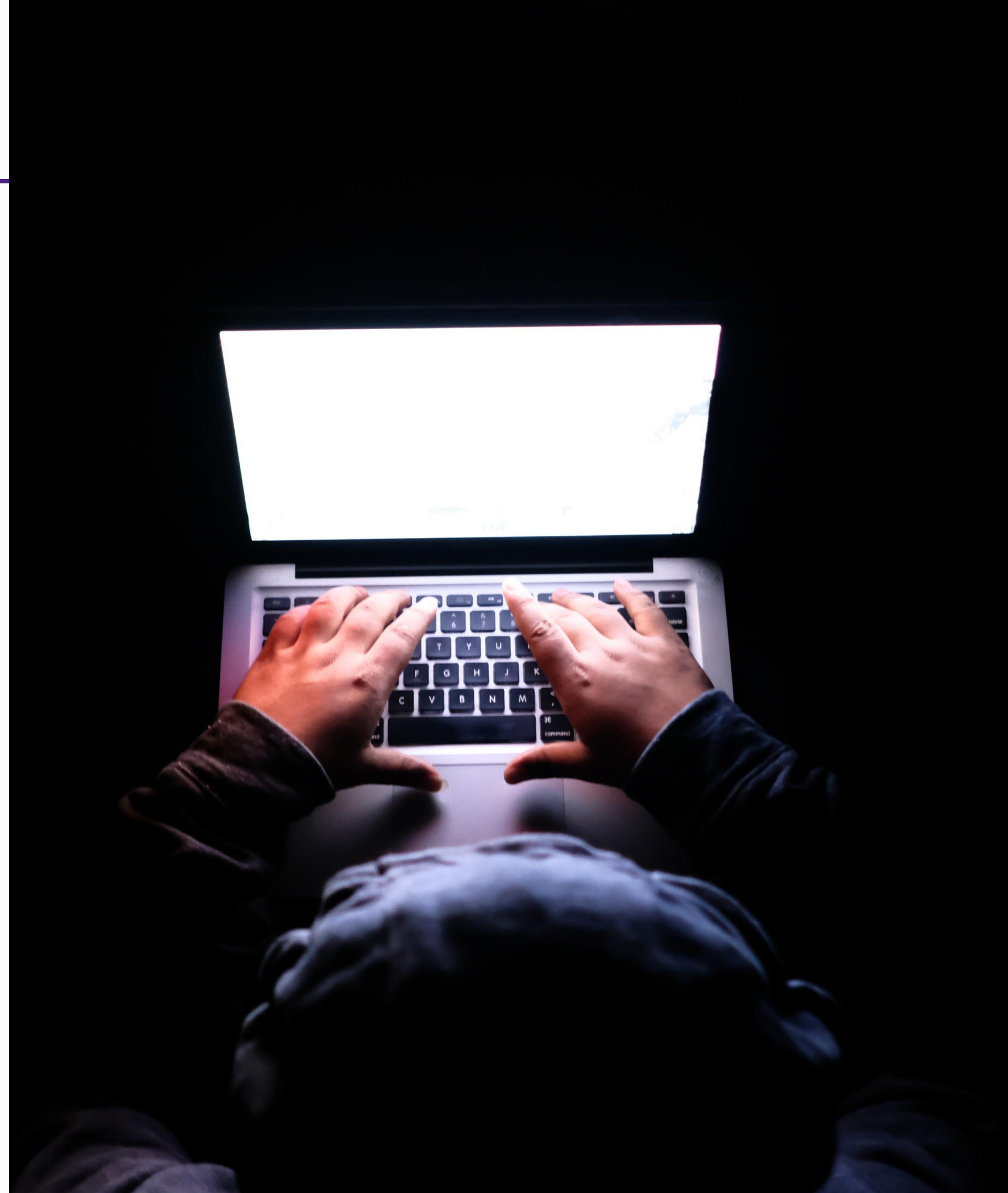


GUESS THE WORD!

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ANSWER!

H A C K I N G



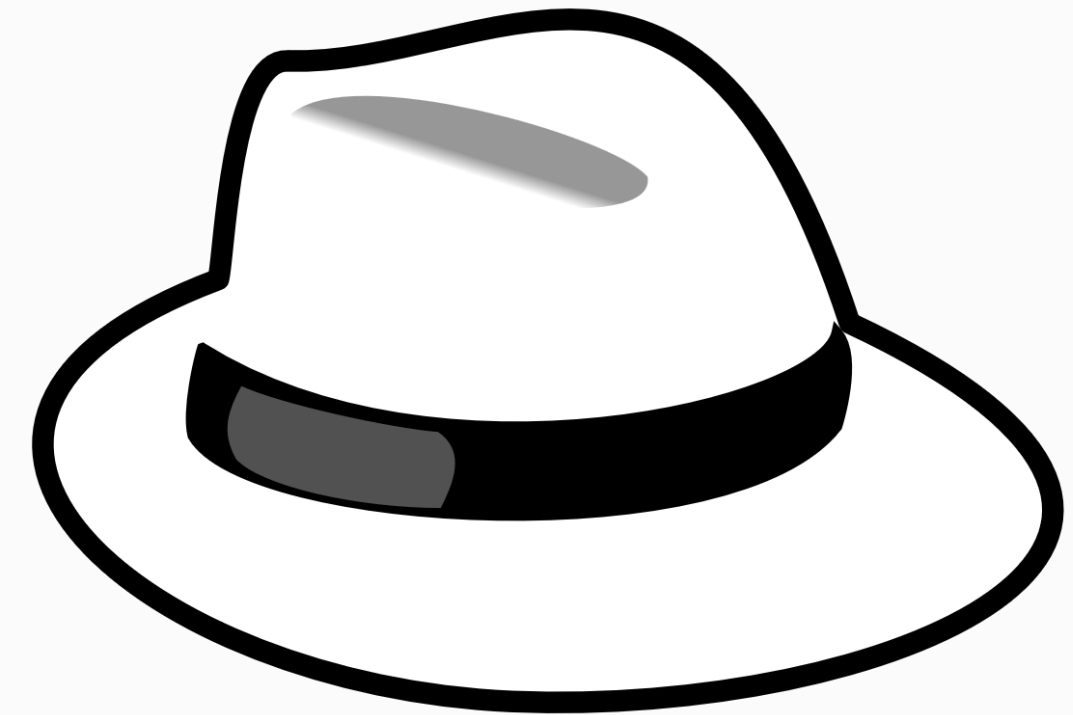
CATEGORIES OF HACKERS

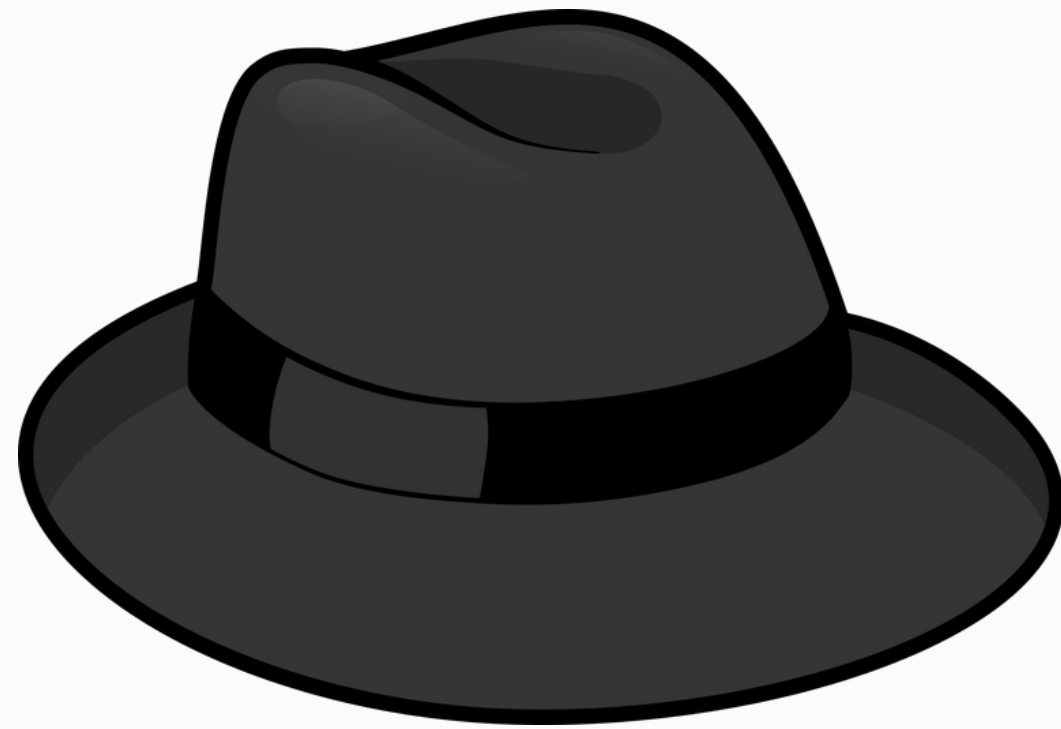
There are different categories of hackers, and the categories are based upon the end goal of the hacker. Some report, some exploit, others do it for fun.



WHITE HAT HACKERS

This set of hackers break into a system, reports vulnerabilities found in the system, and works with the development team to fix the flaws.



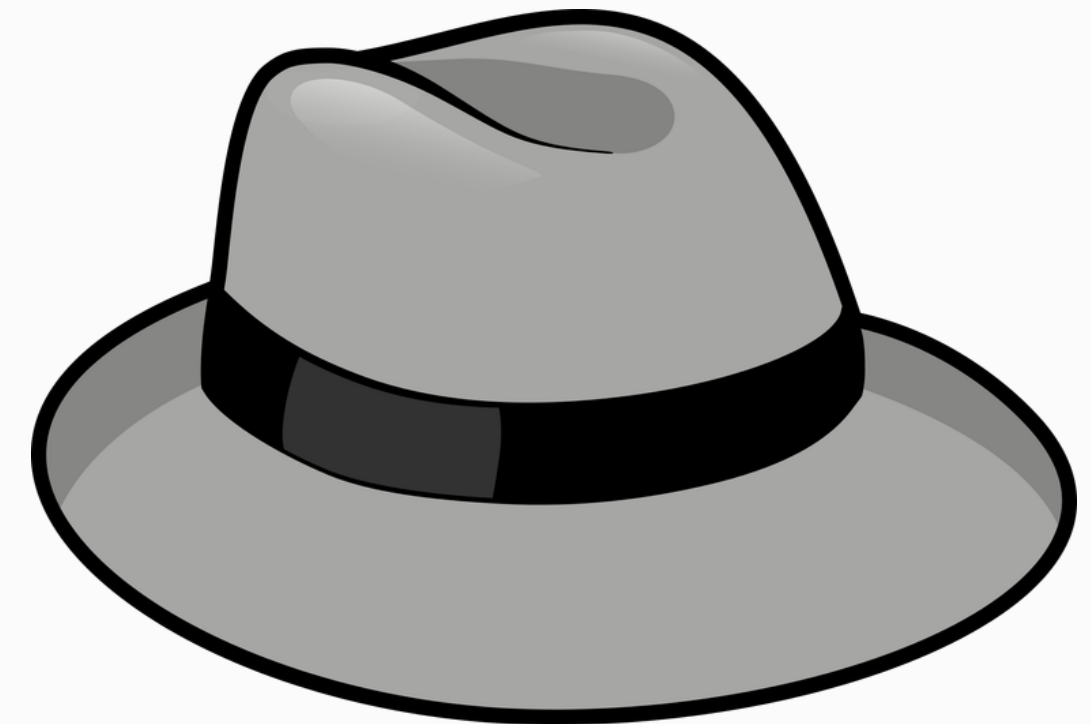


BLACK HAT HACKERS

This set of hackers break into a system, then exploit it for their gain or destroy a company or business. These are the set of people referred to as cybercriminals.

GREY HAT HACKERS

These set of hackers are the hobbyists and people who practice hacking for the fun of it. They break into systems and either report or exploit or even do nothing about it. They have no definite aim, just on the fence between good and evil.





HACKTIVISTS

This set of hackers targets corporate and governmental platforms with intentions to take their systems, expose information, or make their voices heard.



THREAT ACTORS

Threat actors, also known as cyberthreat actors or malicious actors, are individuals or groups that intentionally cause harm to digital devices or systems. Threat actors exploit vulnerabilities in computer systems, networks and software to perpetuate various cyberattacks, including phishing, ransomware and malware attacks.

THE TYPES OF HACKERS AND CYBER ATTACKS IN THE AVIATION INDUSTRY



White hacker or White Unicorn

Authorised or certified hacker who work governments and companies to ensure and protect from malicious cyberattacks



Red hacker

They are also authorised or certified hackers, but they specifically work for the governments, and their main objective is to identify black hackers, cyberterrorists, hacktivists, whistleblowers, and State-nation hackers



Blue hacker

They are much like the white hackers. Although, they are outsourced by firms and working for companies for security testing of their software right before the product launch



Green hacker

They are amateurs in the world of hacking but they are bit different from script kiddies. They care about hacking and strive to become white hackers



Nation sponsored hacker

Are hackers appointed by the government to provide them cybersecurity and to gain confidential information from other countries to stay at the top or to avoid any kind of danger to the country. They are highly paid government workers

Good practices



Black hacker

They reach unauthorised access in companies systems and destroy or steal private data and information, as well as attack vulnerabilities with malicious intent



Nation-state hacker

They can attack for political, business, technological reasons, or as part of a cyberwar



Cyberterrorist

They use cyberattacks as means of furthering their goals, disrupting services, or causing fear. Their main motivations are for political, religious, ideological, and social factors. They attack to endanger national security, cause mass casualties, harm the economy, disrupt public order and morale, and undermine trust in aviation systems



Malicious Insider or Whistleblower

Can be an employee of a company or a government agency with a grudge or a strategic employee who becomes aware of any illegal activities happening within the organisation and can blackmail the organisation for his/her personal gain



The Hacktivist

The hacktivist is a hacker or a group of hackers who gain unauthorised access to government's computer files and networks for further social or political ends



Gray hacker

They are not legally authorised hackers. They work with both good and bad intentions. They can use their skills for personal gain



Script Kiddies

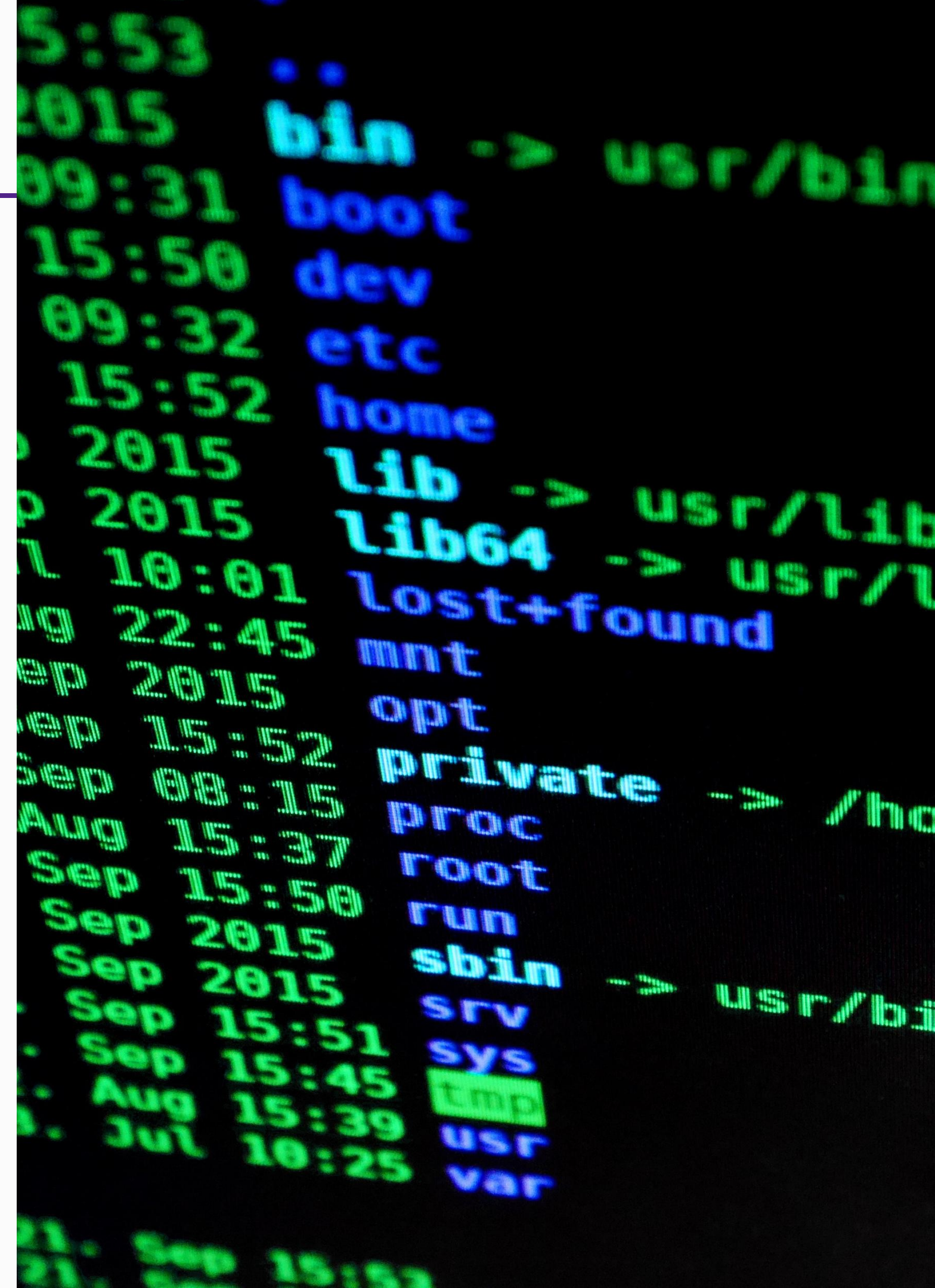
A novice hacker who uses scripts or downloads tools available to attack computer systems and networks and deface websites. His/her main purpose is to impress his/her friends and society

Bad practices

RECAP! RECAP!

Hackers can also use psychology to trick the user into clicking on a malicious attachment or providing personal data. These tactics are referred to as

L O I A C S
N N G E N I E G I E R



RECAP! RECAP!

Hackers can also use psychology to trick the user into clicking on a malicious attachment or providing personal data. These tactics are referred to as

ANSWER!

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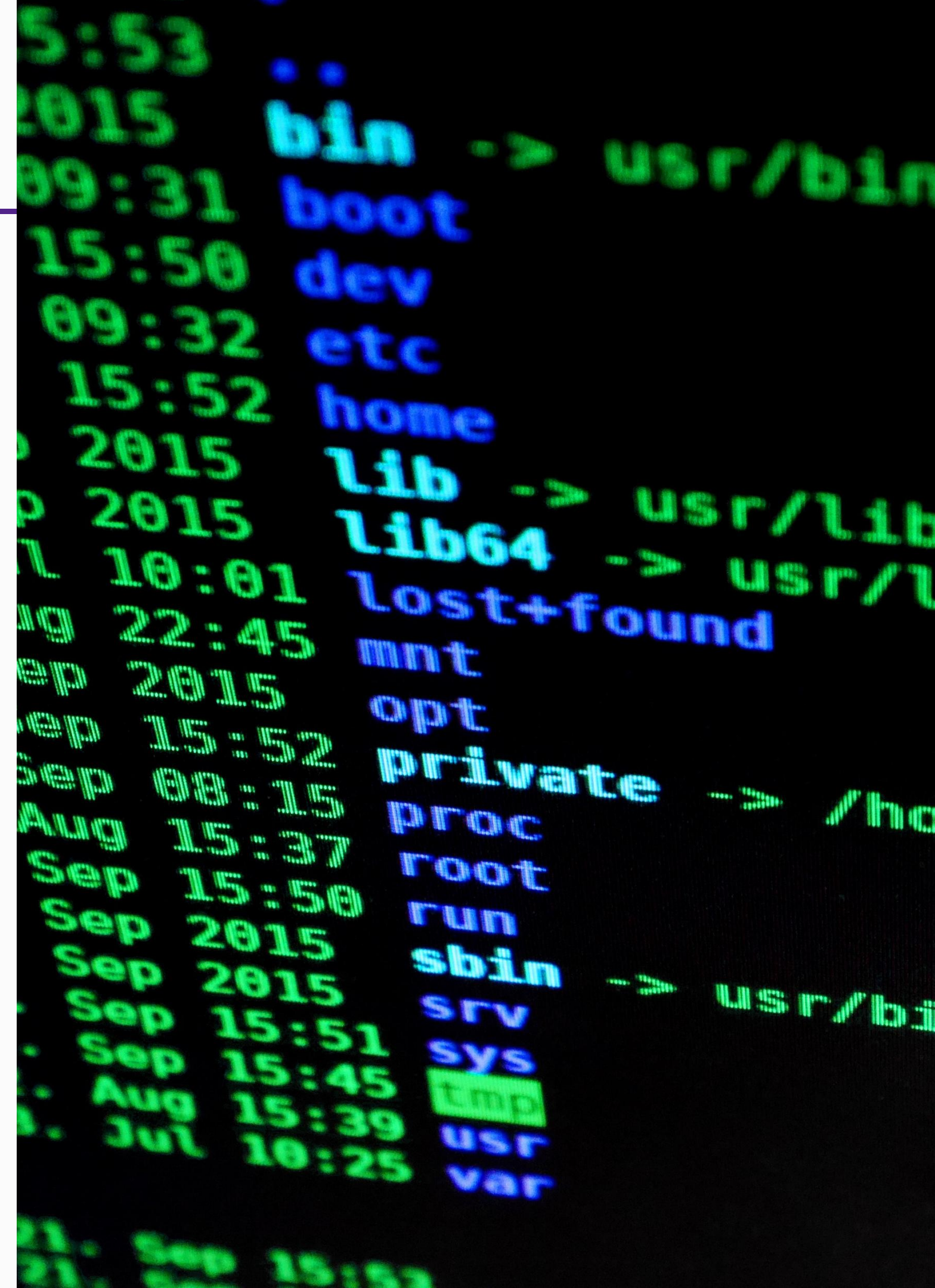
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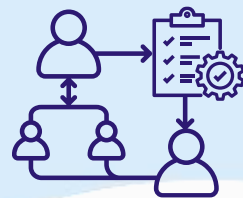


TARGETED SECTORS



Healthcare

This industry is a **primary target** because it stores vast amounts of **sensitive personal and medical data**. Because healthcare services are **critical for saving lives**, cybercriminals believe these organizations are **more likely to pay ransomware demands quickly to restore operations**.



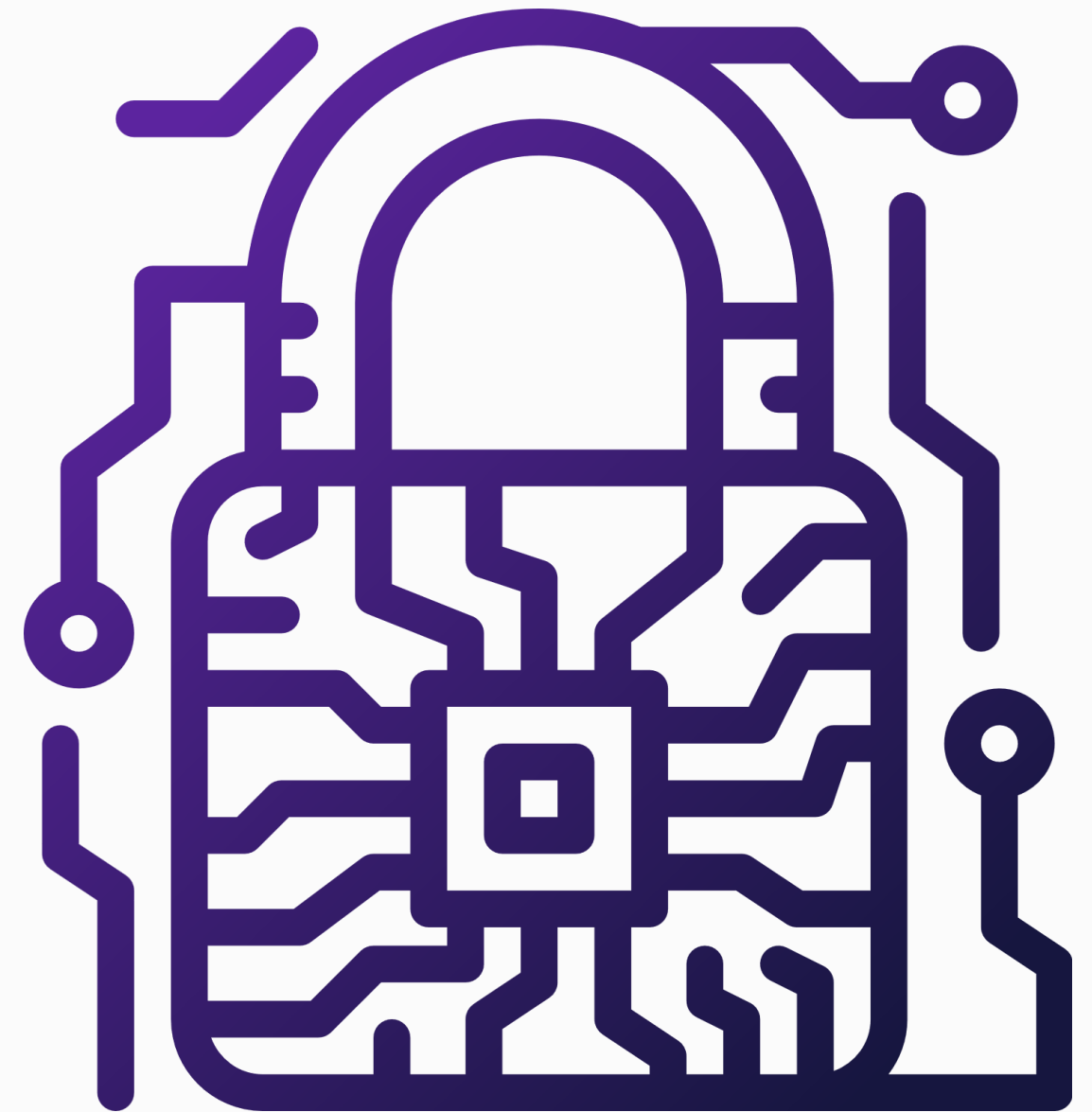
Financial Services

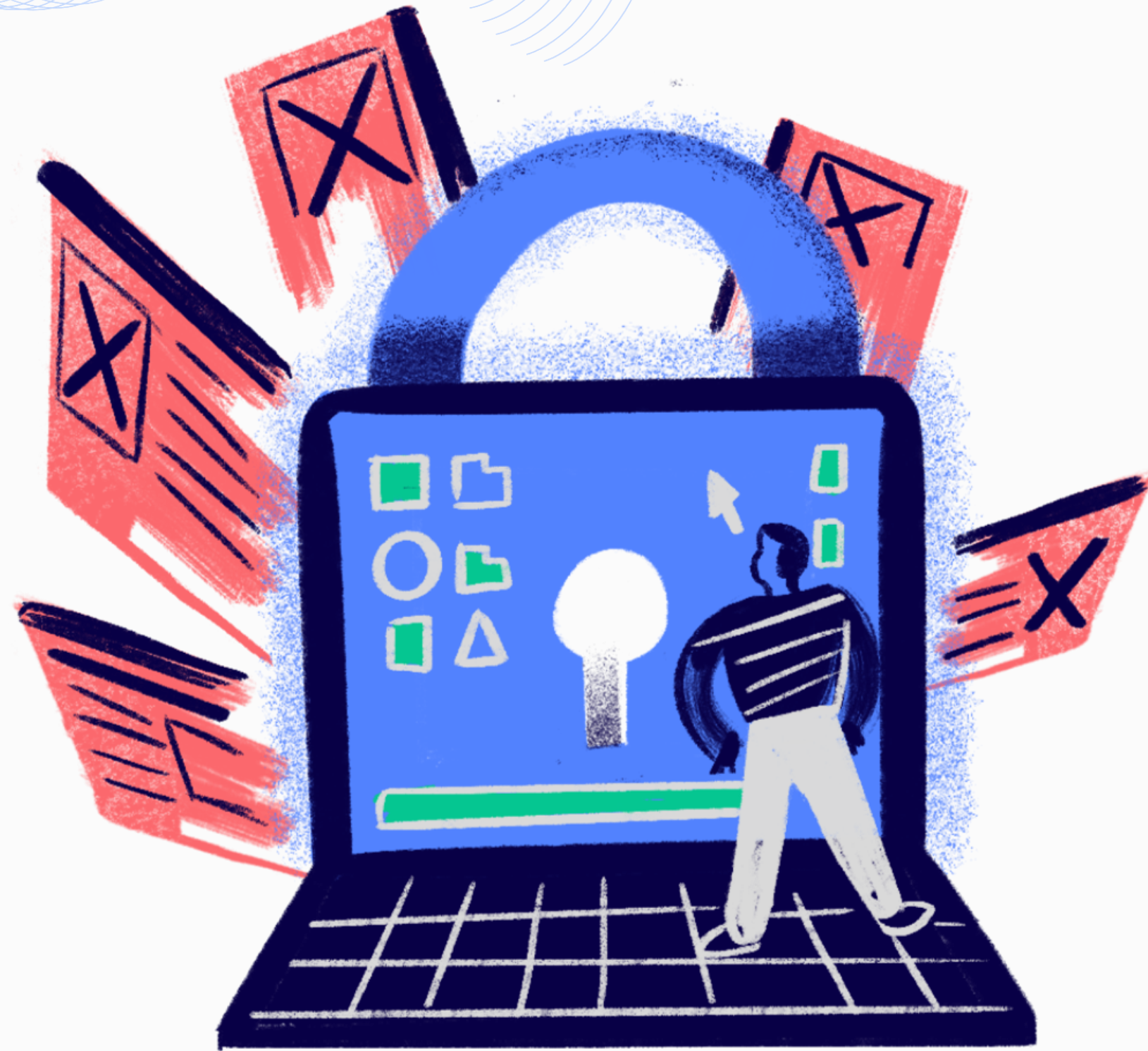
Banks, insurance companies, and investment firms are frequently targeted for direct **financial gain**. Attackers seek to **steal account** credentials, commit fraud, or extort large sums of money through specialized attacks like Corporate Account Takeover (CATO) and ATM Cash Outs.



Government & Public Sector

Hackers target government agencies to **steal confidential intelligence**, personal citizen records (like Social Security numbers), or to disrupt critical infrastructure. These attacks can be part of broader state-sponsored espionage or cyberwarfare activities.





CYBER SECURITY STRATEGIES

Effective security requires
multiple layers of protection

STRONG PASSWORDS

Using complex and
unique passwords
helps prevent
unauthorized access



TWO-FACTOR AUTHENTICATION

An extra security
layer to verify user
identity





FIREWALL AND ANTIVIRUS

Security tools that
monitor and block
suspicious activity

FUTURE OF CYBER SECURITY

Cyber security continues to evolve
with AI and advanced technologies



KEY TAKEAWAYS



Cyber threats are real



Security is essential



Awareness is key



ACTIVITY # 2

CREATE YOUR
GITHUB ACCOUNT
USING YOUR
CYBER HANDLE

Objective

Create a GitHub account using your previously created cybersecurity handle as your username.

Instructions

1. Open your web browser (Chrome, Edge, etc.).
2. Go to GitHub official website
 - Type in the address bar: <https://github.com>
3. Click the “Sign up” button.

Step-by-Step Guide

Step 1: Enter Your Email

- Input your active email address
- Click Continue

Step 2: Create a Password

- Make a strong password
- Click Continue



Step 3: Set Your Username

- Enter your cyber handle from Activity 1
- Click Continue

Step 4: Verification

- Complete the puzzle or verification (captcha)
- Follow instructions on screen

Step 5: Preferences (Optional)

- You may skip or answer questions (e.g., experience level)

Step 6: Verify Your Email

- Open your email inbox
- Look for message from GitHub
- Enter the verification code on the website

IMPORTANT REMINDERS

- Your username must match your cyber handle
- Do NOT use your real name as username
- Save your email and password (you will use this in future activities)

SUBMISSION:

- Submit the following
 - Cyber Handle:
 - GitHub Username:
 - Screenshot of your GitHub profile (after login)